Local Egg Ordering System

A Software for Egg Selling Business

Brumfield, Darliree

Project Summary

Client:	Local small business (The Yolkery)
Prepared by:	Darliree Brumfield
Role:	Full-Stack Developer & System Designer
Date:	March 2026

Purpose:

The Local Egg Ordering System (LEOS) is a web platform designed to help a small local business manage the sale of fresh eggs for curbside pickup. The System allows customers to browse available egg products, submit orders, and receive pickup information while enabling the business owner to manage product availability and customer orders through a simple administrative interface.

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Project Overview

The Local Egg Ordering System (LEOS) is a web platform designed to help a small local farm manage the sale of fresh eggs for curbside pickup. The website allows customers to view available egg products, submit orders, and receive pickup information without needing to manually contact the farm owner.

The primary goal of the platform is to simplify the ordering process for customers while helping the farm owner organize and track incoming orders. The system focuses on simplicity, accessibility, and ease of use, making it suitable for a small-scale local business.

Project Scope

This project focuses on building a lightweight ordering platform for a small egg-selling business. Customers will be able to browse egg products, view descriptions and pricing, and submit orders for local pickup.

The system will include a product listing page, ordering functionality, and basic contact information for the farm owner. The owner will be able to update egg availability and manage incoming orders.

The platform is designed specifically for local curbside pickup and does not include shipping logistics or large-scale e-commerce functionality.

Stakeholders

Table 1. Stakeholders for the Local Egg Ordering System (LEOS)

Stakeholder	Role	Responsibilities
Business Owner (Client)	Backyard egg seller	Manages egg availability, reviews customer orders, and coordinates pickup times
Customers	Local buyers	Browse available eggs, place orders, and pick up eggs at the designated location
System Developer	Developer	Designs, develops, and maintains the Local Egg Ordering System

User Personas

Persona 1: Health-Conscious Consumer

Name: John Doe

User Type: Local customer

John prefers purchasing food from local businesses because he values freshness and transparency in how food is produced. He wants to see information about the eggs and confirm availability before placing an order. His main goal is to easily browse products and place orders for pickup without complicated checkout processes.

Persona 2: Egg Supplier (Client)

Name: Lorraine Gadsden

User Type: Business owner

Lorraine runs a small local egg business and currently manages orders through text messages and phone calls. This can become difficult to organize when multiple customers place orders at the same time. Her goal is to have a simple system where customers can submit orders online while she can easily track them and update egg availability.

Project Modularization — Key Features

1. **Product Catalog Module**
   - Create new egg product listings with name, description, price, and image.
   - Edit existing egg product information.
   - Display available egg products and their details to customers.
   - Delete egg product listing when items are no longer available.
2. **Order Management Module**
   - Receive and store customer order submissions.
   - Confirm order receipt after submission.
   - Allow the owner to view incoming customer orders.
   - Update order status (confirmed, canceled, or on-hold).
   - Send pickup reminders to customers before scheduled pickup.
   - Allow customers to request order rescheduling when necessary.
   - Allow the owner to update or adjust pickup times when needed.
3. **Pickup Information Module**
   - Display pickup location information.
   - Display pickup instructions for customers.
   - Show available pickup hours or pickup days.
   - Notify customers that orders are for local pickup only.
4. **Owner Management Module**
   - Update egg product availability
   - Edit product details
   - View submitted customer orders
   - View customer contact information associated with orders.
   - Track order status using simple labels or status indicators.
5. **Contact and Communication Module**
   - Provide contact form for customer inquiries.
   - Display phone number or email address.

- Provide general business contact information

6. **Informational Content Module**
   - About the business section
   - Business story or background
   - Information about locally raised eggs.
   - Egg freshness and product care/storage recommendations.

Requirements

Functional Requirements

FR1 — Product catalog Module

- FR1.01: The system shall allow the owner to create a new product listing.
- FR1.02: The system shall allow the owner to enter and edit the following product details for each listing: product name, image, description, price, inventory quantity, stock status, and visibility status.
- FR1.03: The system shall allow the owner to delist or permanently delete an existing product listing.
- FR1.04: The system shall automatically mark a product as out of stock when its inventory quantity reaches zero.
- FR1.05: The system shall allow the owner to configure product units such as per dozen, half-dozen, or 18-count.
- FR1.06: The system shall allow the owner to add optional product tags or notes such as “Fresh Today,” “Limited”.
- FR1.07: The system shall display available product listings and product details to customers on the Products Page.
- FR1.08: The system shall hide products marked as not visible from customer view.

FR2 — Order Management Module

- FR2.01 The system shall allow customers to submit an order for one or more egg products.
- FR2.02 The system shall collect customer order information including customer name, contact information, selected products, quantity, and requested pickup time.
- FR2.03 The system shall store submitted customer orders in the system for owner review.
- FR2.04 The system shall display an order confirmation message after a customer successfully submits an order.
- FR2.05 The system shall allow the owner to view all incoming customer orders.
- FR2.06 The system shall allow the owner to update an order’s status as confirmed, canceled, completed, or on hold.
- FR2.07 The system shall allow the owner to update or adjust a customer’s pickup time when necessary.
- FR2.08 The system shall allow customers to request order rescheduling when needed.
- FR2.09 The system shall send a pickup reminder to customers before their scheduled pickup time.
- FR2.10 The system shall associate each order with the customer contact information provided during checkout or submission.

FR3 — Pickup Information Module

- FR3.01 The system shall display the business pickup location to customers.
- FR3.02 The system shall display pickup instructions for customers when placing or reviewing an order.
- FR3.03 The system shall display available pickup days and hours.
- FR3.04 The system shall notify customers that orders are available for local pickup only.
- FR3.05 The system shall display pickup-related updates or notices when provided by the owner.

FR4 — Owner Management Module

- FR4.01 The system shall allow the owner to update product availability.
- FR4.02 The system shall allow the owner to edit product details after a listing has been created.
- FR4.03 The system shall allow the owner to view submitted customer orders.
- FR4.04 The system shall allow the owner to view customer contact information associated with each order.
- FR4.05 The system shall allow the owner to track order status using status labels or indicators.
- FR4.06 The system shall restrict owner management features to authorized administrative access.

FR5 — Contact and Communication Module

- FR5.01 The system shall provide a contact form for customer inquiries.
- FR5.02 The system shall allow customers to submit their name, contact information, and message through the contact form.
- FR5.03 The system shall display the business phone number and/or email address for direct communication.
- FR5.04 The system shall provide general business contact information on the website.

FR6 — Informational Content Module

- FR6.01 The system shall provide an About section describing the business.
- FR6.02 The system shall display the business story or background information.
- FR6.03 The system shall provide information about the locally raised eggs offered by the business.
  FR6.04 The system shall display egg freshness information and care

Nonfunctional Requirements

NFR1 — Performance

- NFR1.01 The system shall load pages and process user actions within 3 seconds under normal operating conditions.

NFR2 — Reliability

- NFR2.01 The system shall maintain periodic backups of order and product data to prevent data loss.

NFR3 — Usability

- NFR3.01 The system shall provide a responsive user interface compatible with both desktop and mobile devices.

NFR4 — Compatibility

- NFR4.01 The system shall function correctly on major modern web browsers including Chrome, Firefox, Safari, and Edge.

NFR5 — Security

- NFR5.01 The system shall protect customer contact information from unauthorized access.

NFR6 — Privacy

- NFR6.01 The system shall clearly state how customer information is collected and used.

NFR7 — Error Handling

- NFR7.01 The system shall display clear and user-friendly error messages when errors occur.

System Design Overview

The Local Egg Ordering System is designed as a web-based application consisting of a user interface and a backend data storage system. The application separates the user experience into two primary layouts: a customer-facing interface and an owner/admin management interface.

The customer-facing interface allows users to browse egg products, place orders, and obtain pickup information. The owner/admin interface allows the farm owner to manage product listings, update availability, and review incoming customer orders.

The website consists of the following primary pages:

- Homepage
- Products Page
- About Page
- Contact / Messaging Page
- Checkout Page

Customer Interaction Flow

Customers interact with the system primarily through the product catalog and checkout process.

To place an order, customers will add desired egg products to their cart. The system requires at least one product in the cart before checkout can begin.

Once the Checkout button is selected, the customer is redirected to the checkout page where the following information must be entered:

- First Name
- Last Name

- Email Address
- Phone Number

Customers may choose one of the following payment options:

- Pay Before Pickup (Cash App or Zelle)
- Pay at Pickup (Cash or mobile payment)

After the order is submitted, the system records the order and provides confirmation to the customer.

Admin Interaction Flow

The owner interface allows the farm owner to manage the business operations through the system.

Using the owner dashboard, the farm owner can:

- Create and edit egg product listings
- Update product availability and pricing
- View incoming customer orders
- View customer contact information associated with orders
- Manage order status for fulfillment

System Architecture Overview

Software Requirements

For the software requirements of this project, the primary operating system that will be utilized is Windows 11 (64-bit). This environment provides compatibility with modern development tools and supports the use of Docker with the WSL2 backend. The main development tools used for this project includes Visual Studio Code, Docker Desktop, PostgreSQL, Redis, FastAPI, Celery, and Node.js with Vite React TypeScript.

Visual Studio Code will serve as the primary development environment and text editor for both backend and frontend development. It will be configured with extensions such as Python and Pylance to assist with linting, formatting, and code navigation. Docker Desktop will be used to manage containerized services, including the PostgreSQL database, Redis message broker, and the FastAPI backend. PostgreSQL will be used as the main relational database for storing application data, while Redis will serve as a message broker to support background task processing with Celery.

On the backend, the project will utilize Python 3.11+ with FastAPI as the primary web framework, Uvicorn as the ASGI server, and Alembic for handling database migrations. The frontend will be developed using Node.js (version 18 or 20) and Vite to build a modern React + TypeScript single-page application. Git will be used for version control, and PowerShell within Windows Terminal will be used to execute setup and deployment commands.

The expected runtime environment will expose services through standard development ports — the FastAPI backend will run on port 8000, the React frontend on port 5173, PostgreSQL on 5432, and Redis on 6379.

Hardware Requirements

This system can be developed on a standard modern development workstation capable of running Docker, Visual Studio Code, and other development dependencies efficiently. A minimum of an Intel i5 or AMD Ryzen 5 processor, 8 gigabytes of RAM, and 20GB of available disk space is required. However, 16 gigabytes of RAM or more is recommended for optimal performance during development.

System Architecture Diagram

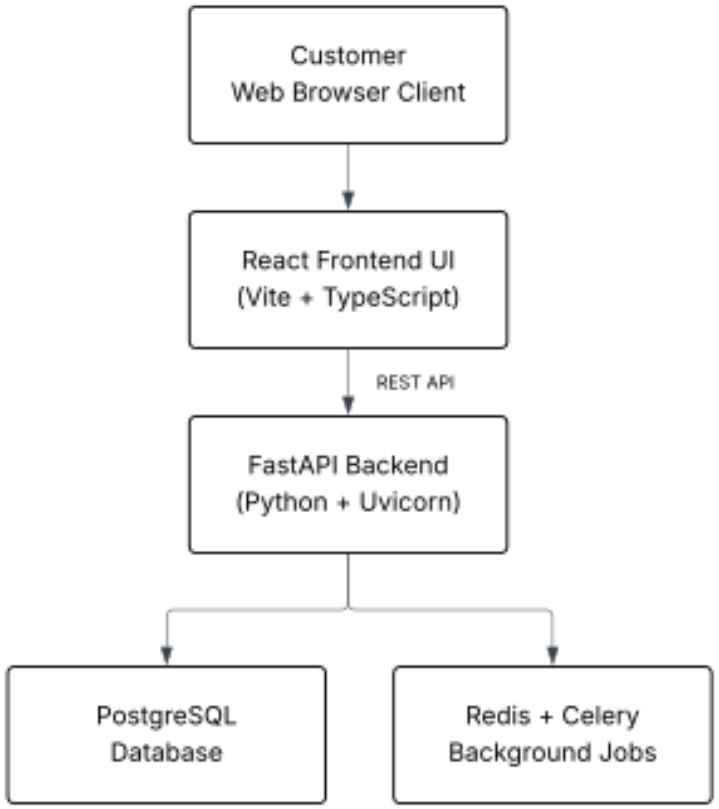


Figure 1. System Architecture Diagram for the Local Egg Ordering System (LEOS)

This diagram illustrates how the React frontend communicates with the FastAPI backend through REST API endpoints. The backend processes requests, stores application data in PostgreSQL, and uses Redis with Celery to support background tasks such as sending order confirmations or pickup reminders.

Site Map

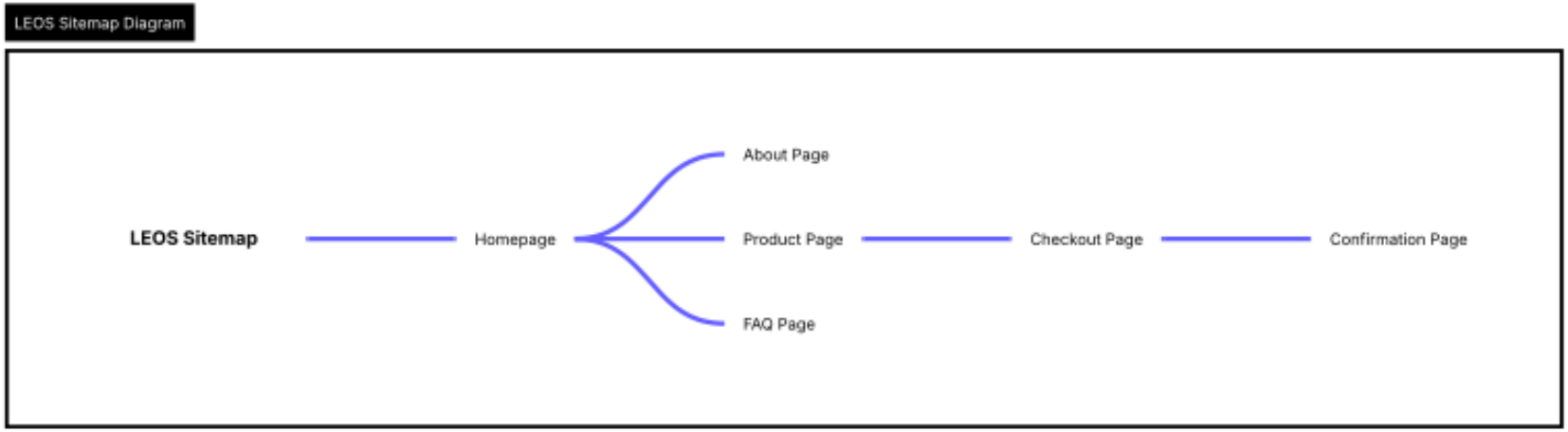


Figure 2. Site Map for the Local Egg Ordering System (LEOS)

The site map illustrates the structural organization of the Local Egg Ordering System website. The homepage acts as the primary entry point for users and provides navigation to the core pages of the platform, including the Products page, Order/Checkout page, About page, FAQ page, and Contact page. This structure ensures that users can easily navigate the website and access key information related to egg products, ordering, and business details.

User Flow

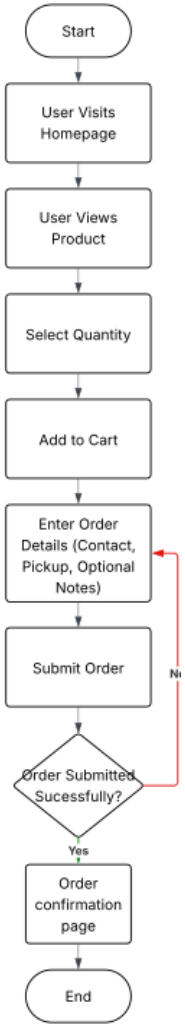


Figure 3. User Flow for the Local Egg Ordering System (LEOS)

Wireframes

The following wireframes illustrate the proposed user interface and layout for the Local Egg Ordering System (LEOS). These visual layouts represent the main pages that users will interact with while browsing egg products, placing orders, and confirming purchases.

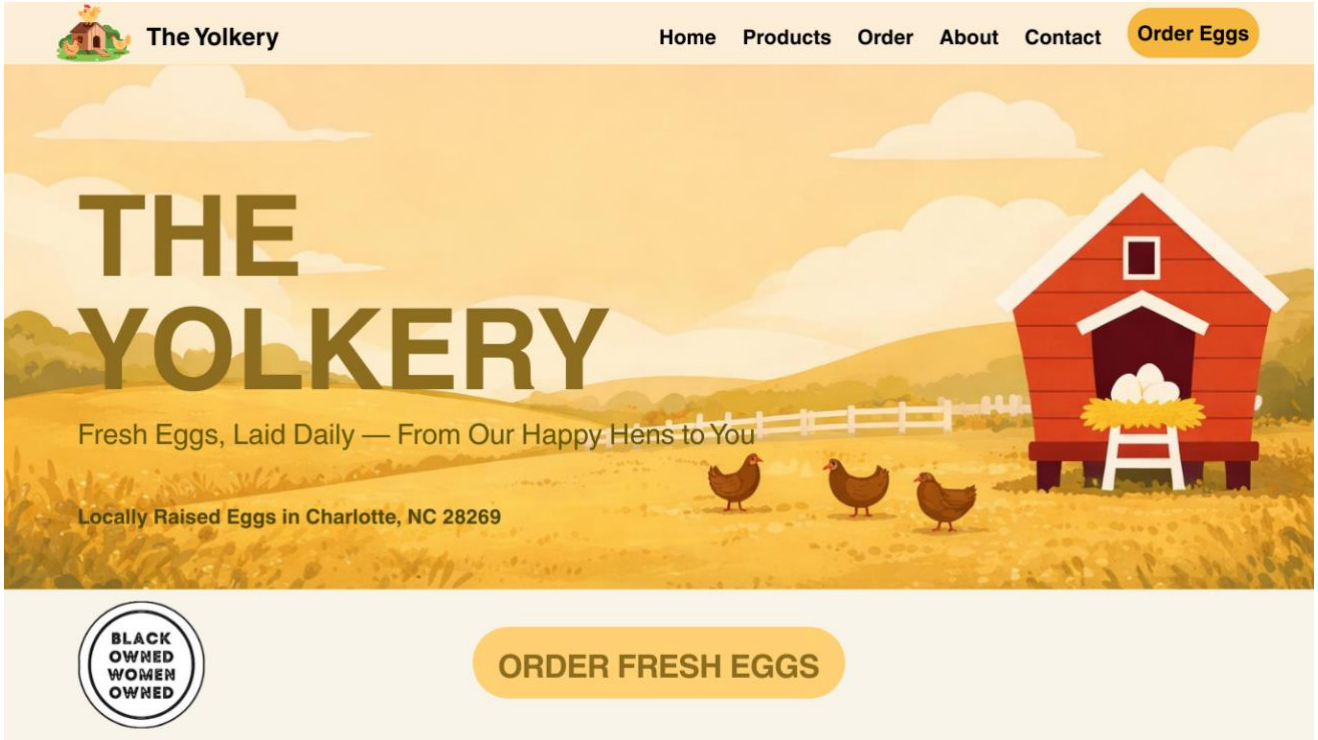
The designs prioritize simplicity, clarity, and accessibility to ensure a smooth ordering experience for customers purchasing locally raised eggs for pickup. The wireframes serve as a structural blueprint for the final user interface and guide the overall user experience of the platform.

Homepage

The homepage introduces visitors to The Yolkery and communicates the business's core value proposition: providing fresh, locally raised eggs to the Charlotte community. The hero section prominently displays the business name, tagline, and a call-to-action button encouraging users to begin the ordering process.

The navigation bar provides access to key pages including Products, Order, About, and Contact. The page also highlights that the business is a Black-owned and woman-owned local business and provides location context without revealing the exact residential pickup address.

Figure 4. Homepage Wireframe

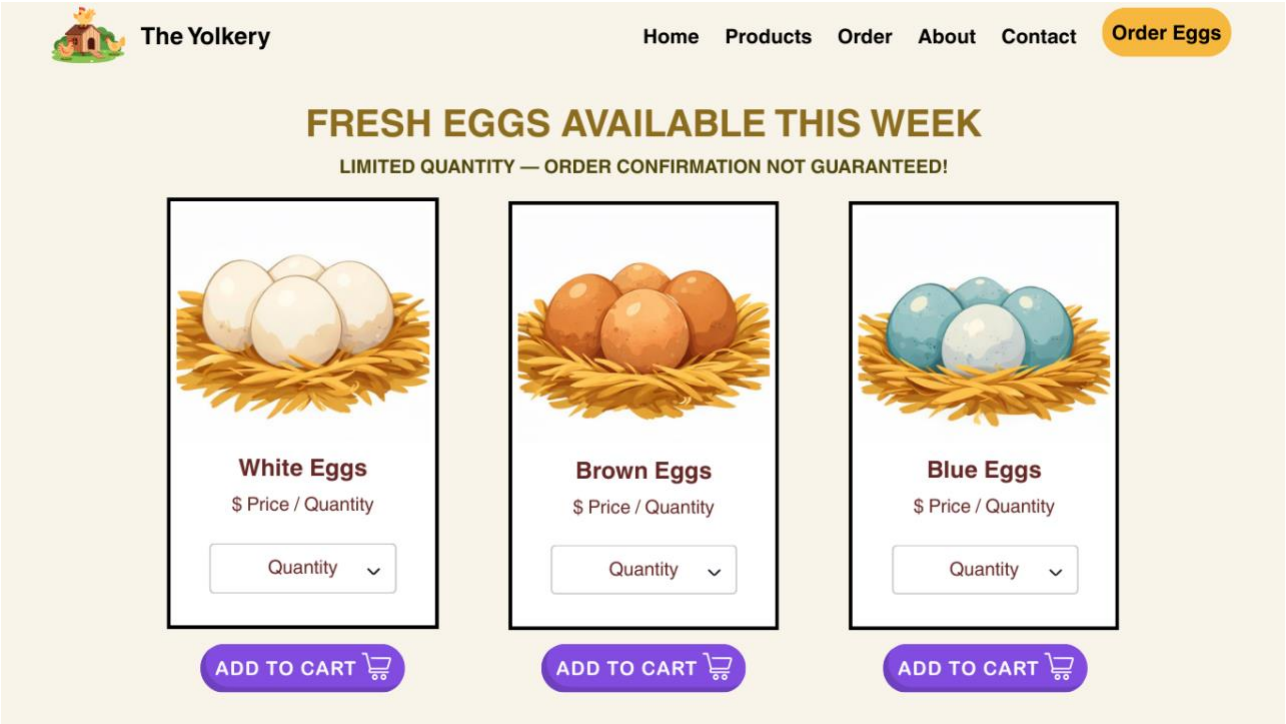


Product Page

The products page displays available egg options using a card-based layout that allows users to easily browse available products. Each product card includes an image, product name, pricing information, and a quantity selection dropdown.

Customers can select the quantity they wish to purchase and add the item to their cart. This layout allows users to compare available egg types and make selections quickly while maintaining a clean and organized interface.

Figure 5. Products Page



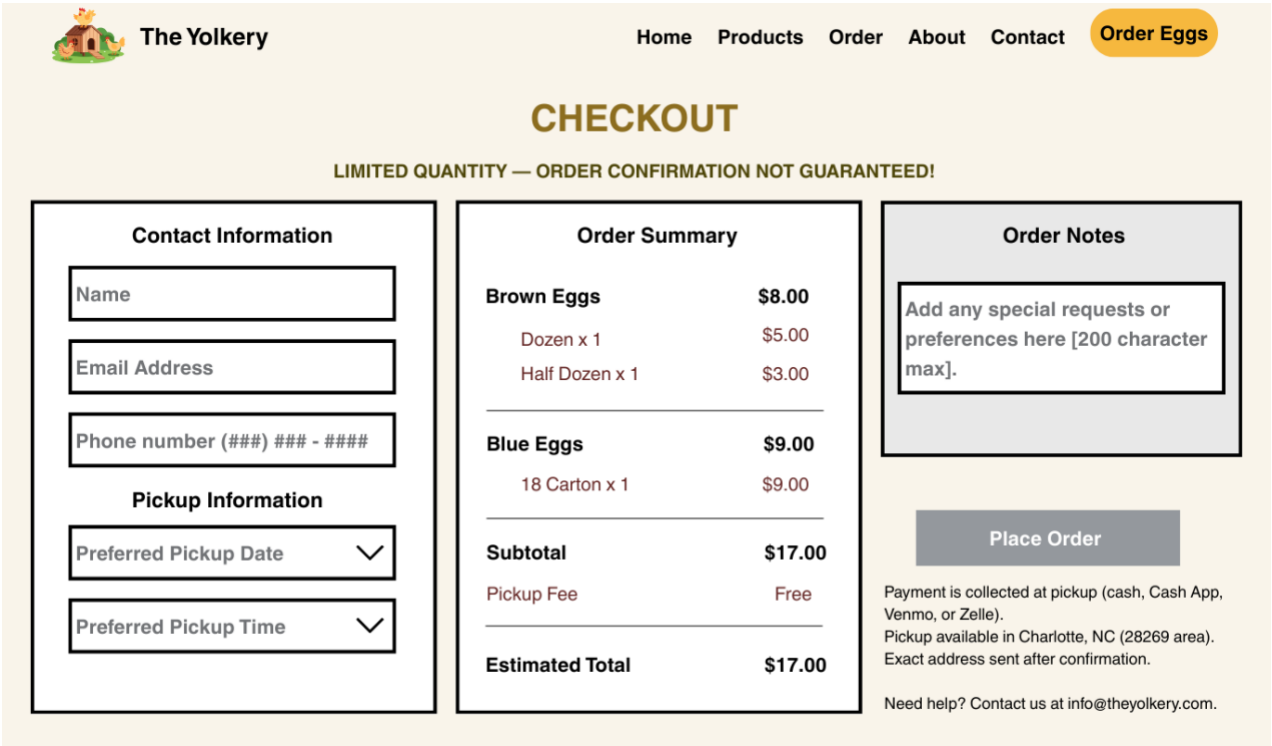
Checkout Page

The checkout page allows customers to finalize their order by providing contact and pickup information. The page includes fields for the customer's name, email address, and phone number, as well as dropdown selections for preferred pickup date and pickup time.

An order summary panel displays the selected products and pricing details, including subtotal and estimated total. Customers may also include optional order notes or special requests.

Important ordering information is displayed on the page, including limited product availability, payment instructions, and pickup location policies.

Figure 6. Checkout Page



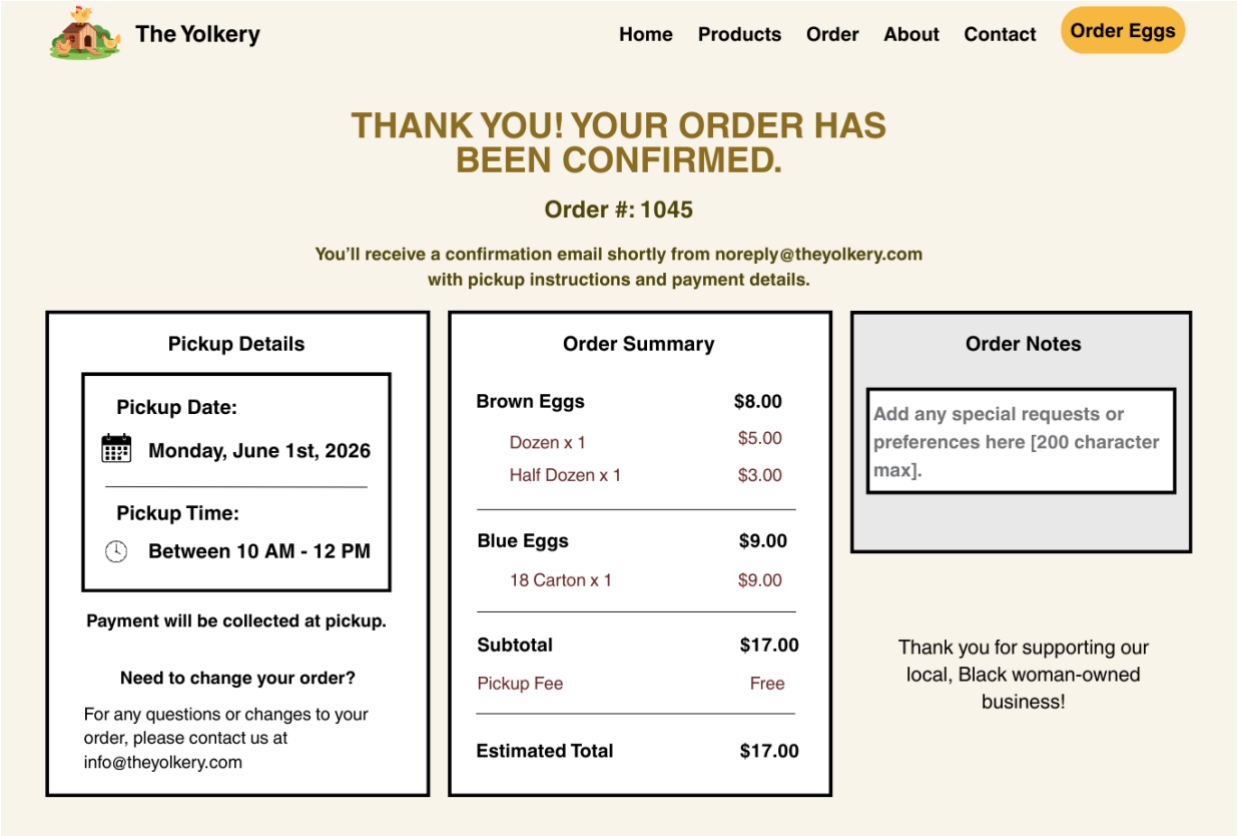
Confirmation Page

After submitting an order, customers are directed to a confirmation page summarizing their purchase. This page confirms that the order request has been received and provides an order number for reference.

The confirmation page displays pickup details, the order summary, and any submitted order notes. Customers are informed that additional pickup instructions and address details will be provided via confirmation email.

The page also provides instructions for contacting the business if the customer needs to change or cancel their order.

Figure 7. Confirmation Page



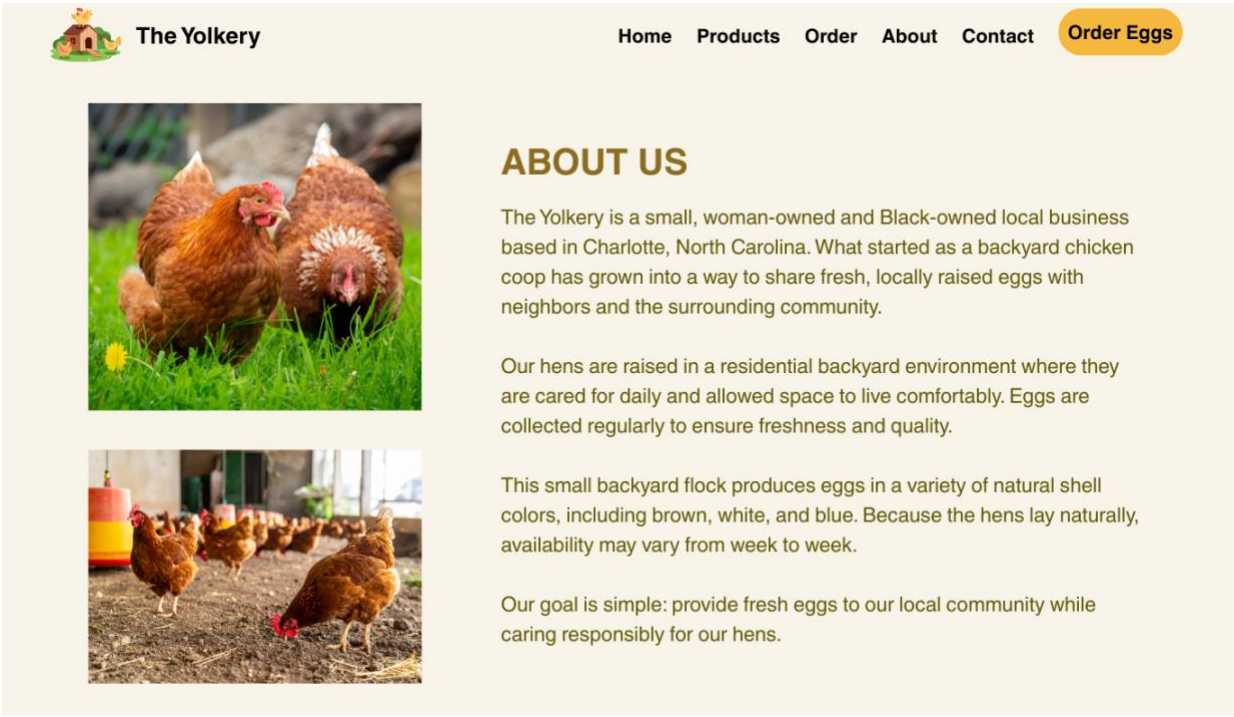
About Page

The About page provides background information about The Yolkery and explains the story behind the business. It highlights that the business is a small, woman-owned and Black-owned backyard egg operation located in Charlotte, North Carolina.

The page includes images of the hens and describes how the chickens are raised and cared for. It also explains that eggs are collected regularly and may vary in shell color depending on the breed of the hens.

This section helps build customer trust and transparency regarding the source of the eggs.

Figure 8. About Page

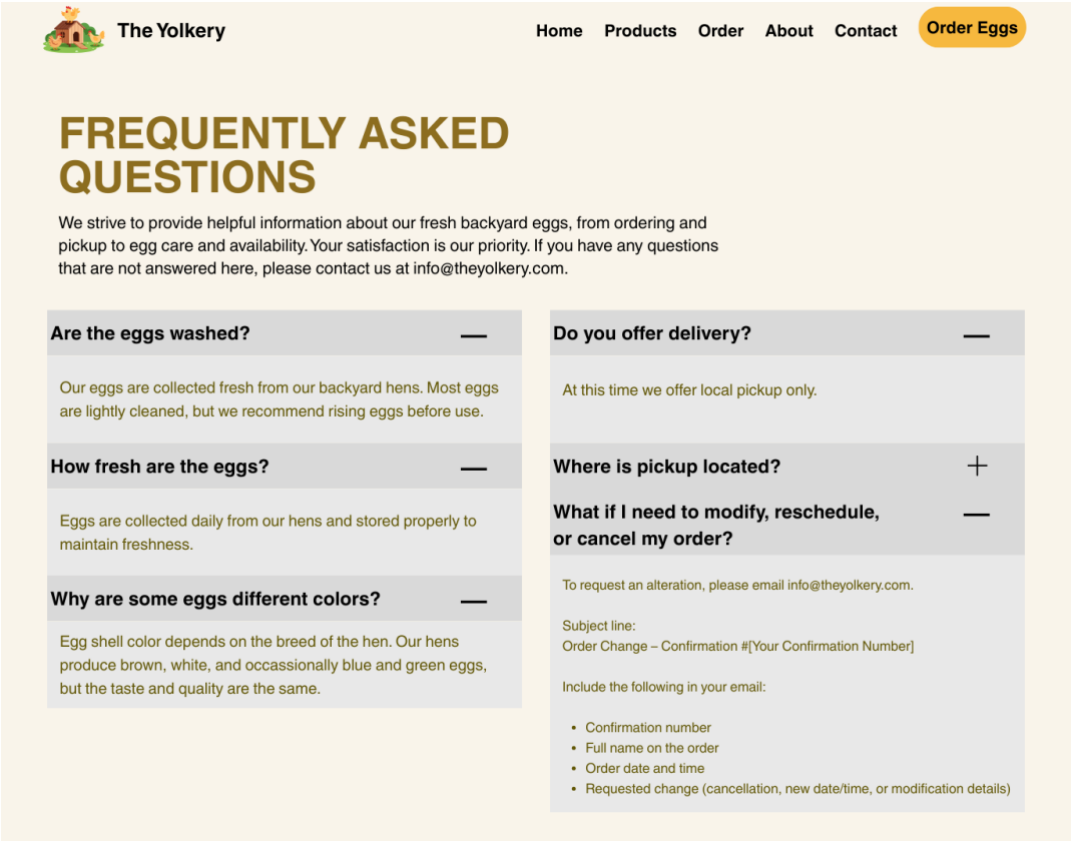


FAQ Page

The FAQ page addresses common customer questions related to egg freshness, pickup logistics, payment methods, and order modifications. The section uses an expandable accordion layout that allows users to quickly view answers while keeping the page visually organized.

This section helps reduce customer confusion and minimizes the need for direct support inquiries by providing clear information about the ordering process.

Figure 9. FAQ Page



Footer Page

The footer provides consistent navigation and contact information across all pages of the website. It includes quick navigation links, business contact information, and the general pickup location area.

The footer also reinforces the business identity as a Black-owned and woman-owned local business while displaying copyright information.

Figure 10. Footer Page

Future Enhancements

Possible future improvements to the Local Egg Ordering System include:

- Online payment integration (Stripe or Square)
- Automated inventory tracking
- SMS order confirmations
- Customer accounts for repeat buyers
- Analytics dashboard for sales trends
- Integration with local farmers market scheduling